

Exact Periods and Zeta Functions of Dyadic Pascal Skew Towers

Route-A structural certificate C166 — internal round 1

Abstract

Let $q = 2^r$ and let T be the d -dimensional affine Pascal tower. We prove that every state has exact period $M = 2^{r + \lceil \log_2 d \rceil}$, so every fixed set is either empty or the full q^d -state space. An explicit binary valuation witness closes the necessity direction. It follows that there are q^d/M primitive cycles and $\zeta_T(z) = (1 - z^M)^{-q^d/M}$. Finite checks are regression sentinels, not the proof. This round leaves the source time-reversal construction to the final internal review.

Keywords: affine dynamics; Pascal coefficients; dyadic valuation; exact period; primitive cycle; Artin–Mazur zeta.

1 Pascal iterate and fixed-point criterion

For $r \geq 1$, $d \geq 2$, and $q = 2^r$, set

$$T(x_1, \dots, x_d) = (x_1 + x_2, \dots, x_{d-1} + x_d, x_d + 1) \pmod{q}.$$

In $R = (\mathbb{Z}/q\mathbb{Z})[t]/(t^{d+1})$, identify x with

$$p_x(t) = 1 + x_d t + x_{d-1} t^2 + \dots + x_1 t^d.$$

The map is multiplication by $1 + t$, hence

$$p_{T^n x} = (1 + t)^n p_x, \text{quad} T^n x = x \iff q \mid \binom{n}{k} \quad (1 \leq k \leq d). \quad (1)$$

The equivalence is state-independent because p_x is a unit.

2 The exact clock

Put $a = \lfloor \log_2 d \rfloor$ and $M = 2^{r+a}$. From

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}$$

we have $v_2(\binom{n}{k}) \geq v_2(n) - v_2(k)$. Thus $M \mid n$ implies that every coefficient in (1) is divisible by q .

For necessity, $q \nmid n$ is detected by $k = 1$. Otherwise suppose $v_2(n) = r + b$ with $0 \leq b < a$. Choose

$$k = 2^{b+1} \leq 2^a \leq d. \quad (2)$$

The low $r + b$ binary digits of $n - 1$ are one, as are the low $b + 1$ digits of $k - 1$. Lucas reduction modulo two gives $\binom{n-1}{k-1} \equiv 1 \pmod{2}$, and therefore

$$v_2\left(\binom{n}{k}\right) = r + b - (b + 1) = r - 1. \quad (3)$$

This violates (1). Hence

$$\text{Fix}(T^n) = \begin{cases} (\mathbb{Z}/q\mathbb{Z})^d, & M \mid n, \\ \emptyset, & M \nmid n. \end{cases} \quad (4)$$

3 Primitive cycles and zeta

Every state has least period M , so there are q^d/M primitive cycles. If $N_n = \#\text{Fix}(T^n)$, then

$$\sum_{n \geq 1} \frac{N_n z^n}{n} = \frac{q^d}{M} \sum_{j \geq 1} \frac{z^{jM}}{j} = -\frac{q^d}{M} \log(1 - z^M).$$

Consequently

$$\zeta_T(z) = (1 - z^M)^{-q^d/M}. \quad (5)$$

The finite Koopman permutation is a direct sum of q^d/M cycles of length M , giving $\det(I - zU_T) = (1 - z^M)^{q^d/M}$.

4 Pivot and boundary

The standalone shear is absorbed by $d = 2$; its odd product-rotation branch is not presented as separate progress. Equations (1)–(5) hold for all source parameters. No complexity theorem, target trace/divisor/counting law, arithmetic local datum, Euler factor, root number, automorphy statement, Hilbert–Pólya construction, or Route-B authorization is claimed. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.

Declarations

All evidence and code are packaged; no external dataset is used. No human or animal subject is involved. No external funding or known competing interest is reported. An AI coding assistant supported planning, drafting, and code review; it was not treated as an external peer reviewer.